Modular proof reduction and guarded dataset labels

2026-11-09. Reviewed symbolic lemmas and bounded software validation.

Layer	Verified condition or remaining obligation
Parameter recognition	Independent low-support basis; n>4r; z=a6=1
Polynomial labels	Exact2 for r1–5, exact3 for r6, interval3–4 above
Retained artifact join	Exact codes, maps, modulus, target and literal rows
Whole dataset admission	Still false: provenance and group lifts are separate

Software: Twenty-three distinct controls pass across inspector, retained join, and capacity-bridge regression. The actual CLI accepts the hand-specified producer-format r1 fixture, labels a rebased r6 input exact3, and rejects a repeated-value selector. Source hashes, theorem bindings and failed preparation are retained. No production artifact, native build or scientific solver was executed.

Basis changes: Full-rank vectors supported below bit k span exactly the k-dimensional low-polynomial space. Invertible input transformations preserve the scoped LFD bounds. FFD2 transfers because literal output rows are reconstructed; output row-span equality alone would not suffice.

Modular correction: For formal $F=p^2+p+s^2$, write F=P Q+R. Actual descent imposes R=0, so former affine chains become L_d=C_d(PQ), generally quadratic in the original coordinates. The no-wrap leading-degree proof cannot discard this correction.

Exact extension: Adding quotient bits with F+P Z=0 forces their unique quadratic definitions. For D>=2, original closure embeds into extended closure, and substituting those definitions maps extended cap D into original cap2D. Consequently padded UNSAT refutation degrees differ by at most a factor two, independent of the number of quotient bits. No unrestricted SAT LFD equivalence or growing lower bound is established.

Evidence: modular-quotient-transport.md and review; no-wrap-basis-recognition-and-label-transfer.md and review; inspector/retained-join source reviews; bound-join-controls/validation.json. The standalone join is not Rust production integration and preserves whole-case admission as false.